

Computing Theoretical Seismograms from a Point Source in a Spherical Multilayered Medium

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Abstract

Computing theoretical seismograms from a point source in a given Earth model is essential for modeling and inversion of observed seismic waveforms for Earth's structure and earthquake source parameters. Here, we derived the propagator matrices and source terms for a spherical multilayered Earth model using the exact earth flattening transformation. We found that their differences from their counterparts in horizontal layered media are inversely proportional to the nondimensional horizontal wavenumber and its higher order. In addition, all the source terms in a spherical layered model have a source-depth dependent scaling factor that differs from in a horizontal layered model by up to 6% for deep earthquakes. The surface displacement produced by a point source can be obtained in a similar form as in horizontal layered media. Computation of theoretical seismograms was implemented using the generalized reflection and transmission coefficients method. Numerical tests show that our formulae and implementation are correct and efficient for computing full-wave seismograms, including the permanent displacements, at teleseismic distances up to 100°. Individual seismic phases can be isolated and analyzed semianalytically because the generalized reflection and transmission method is used. Furthermore, our analytic expression of displacement in terms of the propagator matrices and source terms can be used to derive analytic derivatives of seismograms for full-wave waveform inversion.

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[Supplemental Material](#)

Introduction

Computing theoretical seismograms from a point source in a given Earth model is essential for modeling and inversion of observed seismic waveforms for Earth's structure and earthquake source parameters. Although numerical techniques to simulate wave propagation in 3D Earth structures have been developed in the recent years (e.g., Furumura *et al.*, 1998; Igel *et al.*, 2002; Komatitsch and Tromp, 2002), efficient methods of calculating theoretical seismograms in spherical multilayered Earth models are still useful. This is because that to the first order, the Earth is spherically symmetric and consists of multiple layers. Several spherically symmetric Earth models such as the Preliminary reference Earth model (PREM; Dziewonski and Anderson, 1981) and International Association of Seismology and Physics of the Earth's Interior (IASPEI91; Kennett and Engdahl, 1991) have been established as Earth's reference models. The 1D simplification of Earth model allows derivation of analytic or semianalytic solutions of wave equation, therefore greatly reduces computation time of theoretical seismograms. This is critical for rapid source estimation following a large earthquake (e.g., Dziewonski *et al.*, 1981). Theoretical seismograms of 1D Earth reference models also give us insights to

individual seismic phases in observed seismograms and to identify anomalous velocity structures in the Earth (e.g., Helmberger, 1973; Grand and Helmberger, 1985).

So far, there are several methods to compute full-wave seismograms from a point source in a spherical symmetric Earth model, including the normal-mode summation method (Gilbert, 1971; Masters *et al.*, 2014), the direct solution method (DSM; Geller and Ohminato, 1994; Kawai *et al.*, 2006), the numerical integration method (Friederich and Dalkolmo, 1995), and the spherical Bessel function propagator matrix method (e.g., Gilbert and MacDonald, 1960; Phinney and Alexander, 1966; Wang *et al.*, 2017). Most of the methods use numerical integration for wave propagation in the radial direction in the frequency domain. The computation becomes increasingly challenging as the frequency increases (e.g., Kawai *et al.*, 2006; Yang *et al.*, 2010). Different from the integration

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methods is the earth flattening transformation (EFT) method that uses the Haskell propagator matrix (Haskell, 1953) for wave propagation after transforming the spherical Earth model to a flat Earth model (e.g., Biswas and Knopoff, 1970; Biswas, 1972; Müller, 1985). The numerical integration is replaced by propagator matrix multiplication. There exist many approaches to stabilize the propagator matrix multiplication at high frequencies (Wang and Herrmann, 1980; Kennett, 1983; Luco and Apsel, 1983; Chen, 1993; Wang, 1999). Furthermore, using the Haskell propagator matrix, explicit analytic displacement solution as well as its partial derivatives with respect to the model parameters can be derived in the frequency–wavenumber (f - k) domain (Hu and Zhu, 2017, 2018). The EFT method is also computationally efficient, especially when the epicentral distances are less than 100° , so no core phases are involved.

The traditional EFT method uses the same Haskell propagator matrix of horizontal layered velocity model. The transformation only preserves the travel times of seismic phases. An exact density transformation exists to preserve the amplitudes and shapes of the toroidal waves (Biswas and Knopoff, 1970), but no such exact transformation to density was thought to exist for the spheroidal waves (e.g., Biswas, 1972; Chapman, 1973), and the optimal density transformation for the spheroidal problem was different from the exact toroidal transformation (e.g., Biswas and Knopoff, 1970; Biswas, 1972; Chapman, 1973; Müller, 1985; Herrmann, 2013). Later, Bhattacharya (1996) and Arora *et al.* (1996) found the “exact earth flattening transform” (EEFT) and derived the corresponding Haskell propagator matrix. The EEFT has been used in computing teleseismic surface-wave dispersion (e.g., Arora *et al.*, 1996; Bhattacharya, 1996, 2009; Qureshi and Bhattacharya, 2008). Bhattacharya (2005) used the modified Haskell propagator matrix in a computer code for horizontal layered models to compute theoretical seismograms between 20° and 60° , and showed improvement to the P - SV waveforms. However, he still used the surface harmonic basis functions in the cylindrical coordinates instead of in the spherical coordinates and did not consider changes to the source terms under the transformation, which should affect the amplitude and shape of seismograms (Müller, 1977).

In this study, we followed the approach used by Zhu and Rivera (2002) for horizontal multilayered media to rederive the Haskell propagator matrix in a spherical multilayered model after EEFT. We also derived the source terms of explosion, single force, and double-couple seismic sources. We implemented the formulae in a light-weighted FORTRAN code to calculate theoretical seismograms produced by a point source using the spherical harmonic basis functions and f - k double integration. It computes product of propagator matrices using the generalized reflection and transmission matrix method (Kennett, 1983; Luco and Apsel, 1983; Chen, 1993) to stabilize the computation at high frequency and incorporate the optimization by Pei *et al.* (2008, 2009) for speed. The code, hereinafter

referred to as sFK, is an extension to the popular FK code for horizontal layered velocity models (Zhu and Rivera, 2002). We tested the code using the IASPEI91 model and compared with the DSM method. The results show that sFK is accurate and efficient for both the body and surface waves at teleseismic distances. Numerical analysis and tests also show that the spherical corrections to the source terms can affect the amplitude of seismograms for deep earthquakes up to 6%.

Equations of Motion in a Spherically Symmetrical Earth Model

In the spherical coordinates, the displacement $\mathbf{u}$ and traction σ are expressed as (e.g., Takeuchi and Saito, 1972; Aki and Richards, 1980; Dahlen and Tromp, 1998):

$$\mathbf{u}(r, \theta, \phi, t) = \int e^{-i\omega t} d\omega \sum_{l=0}^{\infty} \sum_{m=-l}^l (U_l^m(r) \mathbf{P}_{lm} + V_l^m(r) \mathbf{B}_{lm} + W_l^m(r) \mathbf{C}_{lm}), \quad (1)$$

$$\sigma(r, \theta, \phi, t) = \int e^{-i\omega t} d\omega \sum_{l=0}^{\infty} \sum_{m=-l}^l (R_l^m(r) \mathbf{P}_{lm} + S_l^m(r) \mathbf{B}_{lm} + T_l^m(r) \mathbf{C}_{lm}), \quad (2)$$

in which

$$\mathbf{P}_{lm} = -Y_l^m \hat{r}, \quad (3)$$

$$\mathbf{B}_{lm} = \frac{1}{\sqrt{l(l+1)}} \left(\frac{\partial Y_l^m}{\partial \theta} \hat{\theta} + \frac{1}{\sin \theta} \frac{\partial Y_l^m}{\partial \phi} \hat{\phi} \right), \quad (4)$$

$$\mathbf{C}_{lm} = \frac{1}{\sqrt{l(l+1)}} \left(\frac{1}{\sin \theta} \frac{\partial Y_l^m}{\partial \phi} \hat{\theta} - \frac{\partial Y_l^m}{\partial \theta} \hat{\phi} \right), \quad (5)$$

are vector spherical harmonics, and

$$Y_l^m(\theta, \phi) = (-1)^m \left[\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!} \right]^{1/2} P_l^m(\cos \theta) e^{im\phi}, \quad (6)$$

is the surface spherical harmonics, and P_l^m is the associate Legendre function.

In an isotropic elastic spherically symmetrical Earth model in which the density ρ and elastic coefficients λ and μ are functions of r only, the second-order differential equation of motion can be reduced a set of first-order ordinary differential equations (e.g., Takeuchi and Saito, 1972; Aki and Richards, 1980; Dahlen and Tromp, 1998):

$$\frac{d\mathbf{b}}{dr} = \mathbf{A}(r)\mathbf{b}, \quad (7)$$

in which $\mathbf{b}$ is the displacement-stress vector, and it is defined as follows:

$$\mathbf{b} = [V_l^m, U_l^m, R_l^m, S_l^m, W_l^m, T_l^m]^T, \quad (8)$$

and

$$\mathbf{A}(r) = \begin{pmatrix} \frac{1}{r} & -\frac{K}{r} & 0 & \frac{1}{\mu} & 0 & 0 \\ \frac{\lambda K}{r(\lambda+2\mu)} & -\frac{2\lambda}{r(\lambda+2\mu)} & \frac{1}{\lambda+2\mu} & 0 & 0 & 0 \\ -\frac{2K\mu(3\lambda+2\mu)}{r^2(\lambda+2\mu)} & \frac{4\mu(3\lambda+2\mu)}{r^2(\lambda+2\mu)}\rho\omega^2 & -\frac{4\mu}{r(\lambda+2\mu)} & \frac{K}{r} & 0 & 0 \\ \frac{4K^2\mu(\lambda+\mu)}{r^2(\lambda+2\mu)} & -\frac{2\mu}{r^2}\rho\omega^2 & -\frac{2K\mu(3\lambda+2\mu)}{r^2(\lambda+2\mu)} & -\frac{K\lambda}{r(\lambda+2\mu)} & -\frac{3}{r} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{r} & \frac{1}{\mu} \\ 0 & 0 & 0 & 0 & \frac{\mu(K^2-2)}{r^2}\rho\omega^2 & -\frac{3}{r} \end{pmatrix}, \quad (9)$$

in which $K = \sqrt{l(l+1)}$ is the nondimensional horizontal wave-number (e.g., [Alterman et al., 1961](#); [Aki and Richards, 1980](#); [Dahlen and Tromp, 1998](#)). $[V_l^m, U_l^m, R_l^m, S_l^m]^T$ is called the spheroidal wavefield, and $[W_l^m, T_l^m]^T$ is called the toroidal field. Equation (9) shows that the two are always decoupled in an isotropic spherical symmetric earth model.

EEFT and Propagator Matrix

If the velocities and density functions of the model take the following forms:

$$\alpha(r) = (r/a)\alpha_0, \quad (10)$$

$$\beta(r) = (r/a)\beta_0, \quad (11)$$

$$\rho(r) = (r/a)^{-4}\rho_0, \quad (12)$$

in which a is the radius of earth, we define an auxiliary displacement-stress vector as follows:

$$\mathbf{y} = a^2 \sqrt{\frac{4\pi}{2l+1}} \left(\frac{r}{a}\right)^{-\frac{1}{2}} [V_l^m, U_l^m, \frac{a}{K} \left(\frac{r}{a}\right)^3 R_l^m, \frac{a}{K} \left(\frac{r}{a}\right)^3 S_l^m, W_l^m, \frac{a}{K} \left(\frac{r}{a}\right)^3 T_l^m]^T, \quad (13)$$

and introduce a nondimensional depth variable:

$$z = \ln\left(\frac{r}{a}\right). \quad (14)$$

Equations (10)–(12) and equation (14) are referred as the EEFT (e.g., [Bhattacharya, 2005](#)). Under the EEFT, equation (7) can be rewritten as follows:

$$\frac{d\mathbf{y}}{dz} = \mathbf{M}\mathbf{y}, \quad (15)$$

and $\mathbf{M}$ is a constant matrix independent of z . For the spheroid motion:

$$\mathbf{M} = K \begin{pmatrix} \frac{1}{2K} & -1 & 0 & \frac{1}{\mu_0} \\ 1-2\xi & -\frac{5-8\xi}{2K} & \frac{\xi}{\mu_0} & 0 \\ \frac{2\mu_0(4\xi-3)}{K} & -\rho_0 \left(\frac{a\omega}{K}\right)^2 + \frac{4\mu_0(\pi)}{K^2} & \frac{5-8\xi}{2K} & 1 \\ 4\mu_0\xi_1 - \rho_0 \left(\frac{a\omega}{K}\right)^2 - \frac{2\mu_0}{K^2} & \frac{2\mu_0(4\xi-3)}{K} & 2\xi-1 & -\frac{1}{2K} \end{pmatrix}, \quad (16)$$

in which, $\xi = \mu_0/(\lambda_0 + 2\mu_0)$, and $\xi_1 = 1 - \xi$. Subscript 0 of model parameters will be dropped hereafter without losing clarity. $\mathbf{M}$ becomes the same as its counterpart in the horizontal layered model (e.g., [Zhu and Rivera, 2002](#)) when K becomes infinite.

The general solution to equation (15) is as follows:

$$\mathbf{y}(z) = \mathbf{a}(z - z_0)\mathbf{y}(z_0), \quad (17)$$

in which $\mathbf{a}(z - z_0) = e^{(z-z_0)\mathbf{M}}$ is called the Haskell propagator matrix. Following [Zhu and Rivera \(2002\)](#), we used the Jordan decomposition of $\mathbf{M}$:

$$\mathbf{M} = \mathbf{E}\mathbf{J}\mathbf{E}^{-1}, \quad (18)$$

to find the propagator matrix:

$$\mathbf{a}(z - z_0) = \mathbf{E}e^{(z-z_0)\mathbf{J}}\mathbf{E}^{-1}. \quad (19)$$

The explicit expressions of $\mathbf{J}$, $\mathbf{E}$, $\mathbf{E}^{-1}$, and propagator matrix $\mathbf{a}$ of the spheroidal and toroidal problems are given in [Appendices A](#) and [B](#), respectively. Compared with their counterparts in horizontal layered media ([Zhu and Rivera, 2002](#)), the newly derived propagator matrices in spherical layered media have extra terms that decrease with the increase of K . The differences between the two vanish when K becomes infinite.

The source terms

The displacement-stress vector jumps caused by different seismic sources (the source terms) were derived by several authors (e.g., [Saito, 1967](#); [Takeuchi and Saito, 1972](#); [Chapman and Orcutt, 1985](#)). We used the results of [Takeuchi and Saito \(1972\)](#) to find the jumps to our auxiliary stress-displacement vector by taking into consideration of the differences in the definitions of the stress-displacement vector and vector spherical harmonics.

Explosion source

Only the $m = 0$ term exists for the isotropic source. Using equation (266) in [Takeuchi and Saito \(1972\)](#), we first obtained the jump to our stress-displacement vector at the source level r_s :

$$\Delta \mathbf{b} = \sqrt{\frac{2l+1}{4\pi}} \frac{1}{r_s^2} \left[0, \frac{\xi}{\mu}, -\frac{4\xi}{r_s}, \frac{2\xi K}{r_s}, 0, 0 \right]^T. \quad (20)$$

We then converted it to the jump of our auxiliary displacement-stress vector using equation (13) and transforming the medium parameters using equations (10)–(12):

$$\mathbf{s}^0 = \sqrt{\frac{a}{r_s}} \left[0, \frac{\xi}{\mu}, -\frac{4\xi}{K}, 2\xi, 0, 0 \right]^T. \quad (21)$$

Single force

There are three nonzero source terms ($m = 0, \pm 1$) for the single force source. The terms for $m = \pm 1$ can be combined into one:

$$\mathbf{s}^0 = \sqrt{\frac{r_s}{a}} \frac{a}{K} [0, 0, -1, 0, 0, 0]^T, \quad (22)$$

$$\mathbf{s}^1 = \sqrt{\frac{r_s}{a}} \frac{a}{K} [0, 0, 0, -1, 0, 1]^T. \quad (23)$$

Double-couple without torque

Similarly, the five nonzero source terms ($m = 0, \pm 1, \pm 2$) can be reduced to three:

$$\mathbf{s}^0 = \sqrt{\frac{a}{r_s}} \left[0, \frac{2\xi}{\mu}, -\frac{2(4\xi-3)}{K}, 4\xi-3, 0, 0 \right]^T, \quad (24)$$

$$\mathbf{s}^1 = \sqrt{\frac{a}{r_s}} \left[\frac{1}{\mu}, 0, 0, 0, -\frac{1}{\mu}, 0 \right]^T, \quad (25)$$

$$\mathbf{s}^2 = \sqrt{\frac{a}{r_s}} \frac{\sqrt{K^2-2}}{K} [0, 0, 0, 1, 0, -1]^T. \quad (26)$$

Compared with the source terms in horizontal layered media (Zhu and Rivera, 2002), the spherical source terms have two noticeable differences due to the geometry difference between the spherical and cylindrical coordinates: (1) the $m = 0$ source terms have a nonzero jump in the vertical stress component, $-\frac{4\xi}{K}$ for the explosion source, and $-\frac{2(4\xi-3)}{K}$ for the double-couple source. It decreases as K increases, but it is not negligible for $K < 100$ using $\xi \approx \frac{1}{3}$ of typical rocks; (2) all source terms have a source depth h -dependent scaling factor, $\sqrt{\frac{a}{r_s}}$ for the explosion and double couple sources, and $\sqrt{\frac{r_s}{a}}$ for the single force source. The scaling factor deviates from unity by $\sim \frac{h}{2a}$. So for a deep earthquake at 700 km depth,

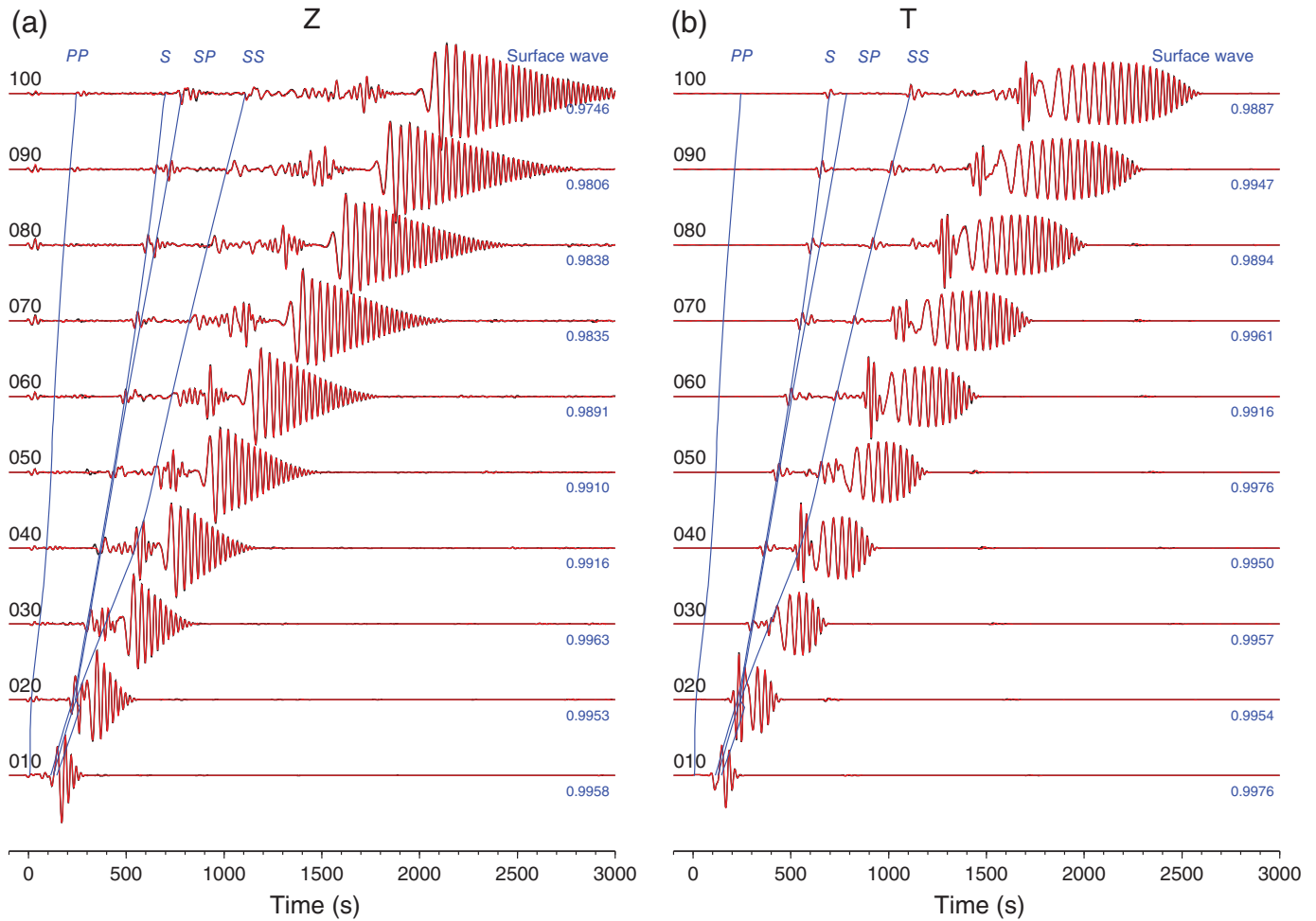
the difference in seismogram amplitudes predicted using the source terms in horizontal and spherical layered models can be as large as 6%.

Implementation and numerical tests

The derived propagator matrices and source terms lead to analytic expression of surface displacement in the f - k domain from a point source in a multilayered model, see Zhu and Rivera (2002) for details. Theoretical seismograms can then be calculated using equation (1) by a wavenumber summation and inverse Fourier transform. However, multiplication of the Haskell propagator matrices of different layers is numerically difficult due to loss of significant digits at high frequencies. Various methods exist to mitigate the problem, including the compound matrix method (e.g., Dunkin, 1965; Watson, 1970; Wang and Herrmann, 1980), the reflectivity method (Fuchs and Müller, 1971; Kennett, 1983), and the orthonormalized propagator method (Chapman and Phinney, 1972; Wang, 1999). Here, we used the generalized reflection and transmission method (Luco and Apsel, 1983; Chen, 1993), which eliminates the growing exponential terms and is stable at high frequencies. We also speeded up the computation using a fast algorithm of Pei *et al.* (2008, 2009) and parallel computation with openMP. These procedures were implemented in a FORTRAN computer code named sFK.

To verify the correctness of the implementation, we tested sFK using the IASPEI91 model down to the upper part of the liquid outer core. We computed seismograms from a 100 km deep event at epicentral distances from 10° to 100° in a frequency range of 0.002–0.04 Hz. The seismograms were compared with those using the DSM method (Fig. 1). The arrival times of several phases in the waveforms can be clearly identified (Fig. 1). We also calculated the cross-correlation coefficients between the sFK and DSM synthetic seismograms. The results show that the sFK and DSM synthetic seismograms have the cross-correlation coefficients above 0.97 for P - SV components and above 0.98 for SH components, and the waveforms agree with each other in both the body and surface waves (Fig. 1).

Compared with other methods such as the DSM, the sFK method is more efficient to compute full-wave seismograms at teleseismic distances up to 100° , because only the Earth model above the core is needed. For the above tests, the computation times of the parallel version of the sFK (with eight central processing units), the serial version of the sFK, and the serial version of the DSM code are 59, 320, and 371 s, respectively. The computation time is proportional to the number of layers in the model. In the sFK computation, we discretized the model in 89 layers. The accuracy of the sFK seismograms at high frequencies could be improved if the model is discretized in more layers, but computation time would increase. The appropriate model discretization is related to both the frequency band and the required accuracy of the synthetic seismograms. The “optimum”



model discretization needs to be determined for each application. In this study, we consider the effect of model discretization on the accuracy of the synthetics by numerical tests. In detail, assuming the “exact” synthetic seismograms were obtained using a 1600-layer model, several choice of model discretization (number of layers ranging from 40 to 800) were compared with the “exact” synthetics in different frequency bands (Fig. S1, available in the supplemental material to this article). It is shown that the root mean square error (rmse) decreases because the increase of number of layers of the models, and the finer model discretization is required for synthetic seismograms with higher frequencies (Fig. S1). In our test, it seems that a 200-layer model is sufficient for calculation of accurate synthetic seismograms with periods longer than 5 s (rmse smaller than 0.1%).

Discussion and Conclusions

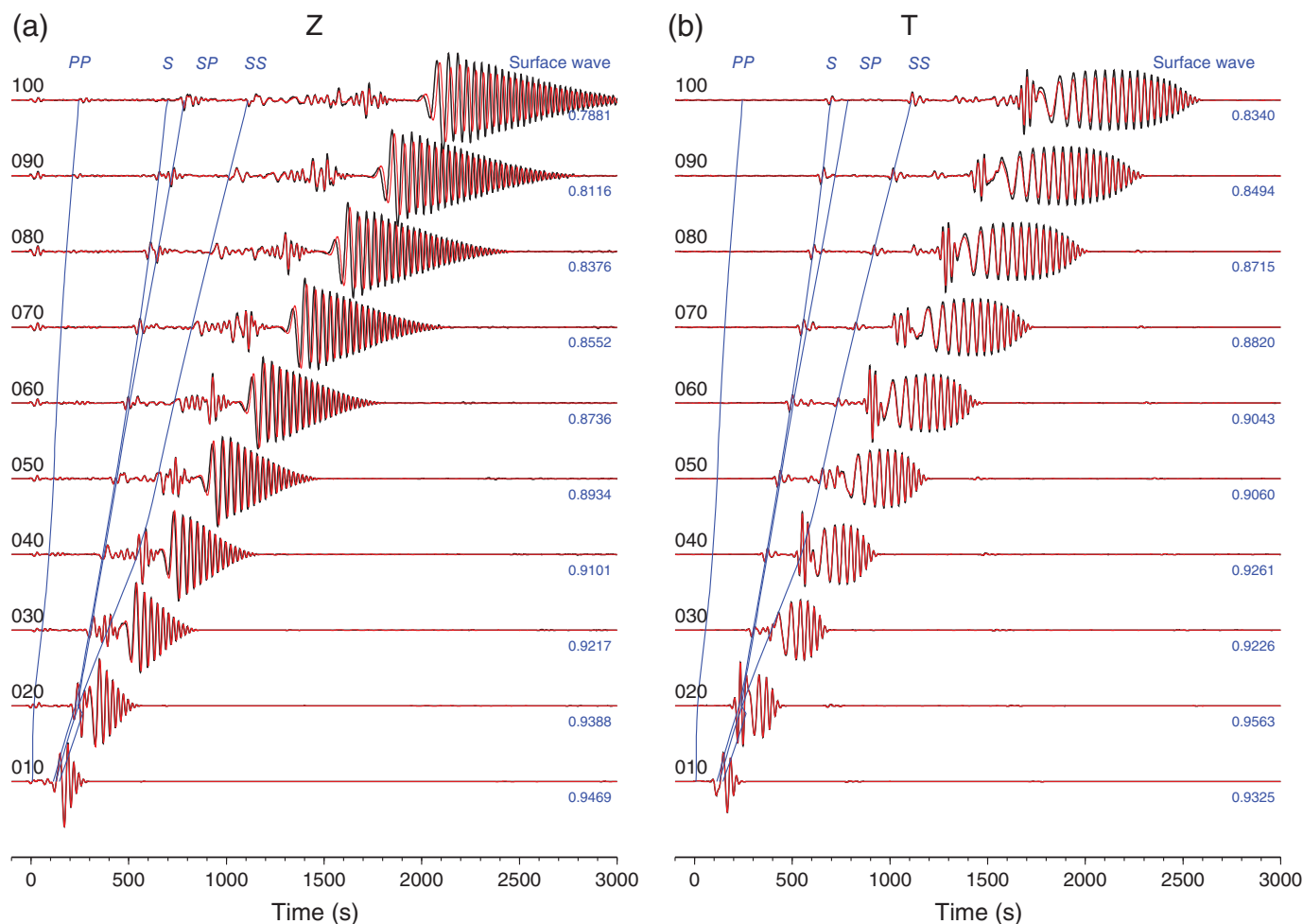
We showed that at large K and $\frac{h}{a} \ll 1$, our derived propagator matrices and source terms become the same as their counterparts in horizontal layered media in [Zhu and Rivera \(2002\)](#). Consequently, the auxiliary displacement-stress vector $\mathbf{y}$ solution at the surface becomes the same as its counterpart in horizontal layered media. Furthermore, at large K , the associate Legendre function is approximate to the Bessel function (e.g., [Olver et al., 2010](#)):

Figure 1. Comparison of direct solution method (DSM; black) with sFK (red) seismograms from 10° to 100° from a 100 km deep earthquake in the International Association of Seismology and Physics of the Earth’s Interior (IASPEI91) model. The magnitude of the earthquake is M_w 5.94, with a source duration of 8 s and strike 39°, dip 58°, and rake 72°. A band-pass filter of 0.002–0.04 Hz was applied to the seismograms. The seismograms were aligned according to the first P arrival time at 0 s. The arrival times of several phases (PP , S , SP , and SS) were plotted and labeled. The epicentral distances (in black) and the cross-correlation coefficients (in blue) were labeled next to each trace. Panels (a) and (b) are the vertical (Z)- and transverse (T)- component seismograms, respectively.

$$\lim_{l \rightarrow \infty} (-1)^m \left[\frac{(l-m)!}{(l+m)!} \right]^{1/2} P_l^m(\cos \theta) = \sqrt{\frac{\theta}{\sin \theta}} J_m \left(\left(l + \frac{1}{2} \right) \theta \right), \quad (27)$$

Therefore, equation (1) becomes:

$$\mathbf{u}(r', \theta', z', t) \simeq \frac{1}{2\pi} \sqrt{\frac{\theta}{\sin \theta}} \int e^{-i\omega t} d\omega \sum_{l=0}^{\infty} \frac{l+1/2}{a^2} \sum_{m=-l}^l (y_2(r) \mathbf{P}'_{lm} + y_1(r) \mathbf{B}'_{lm} + y_5(r) \mathbf{C}'_{lm}), \quad (28)$$



in which $\mathbf{P}'_{lm}$, $\mathbf{B}'_{lm}$, and $\mathbf{C}'_{lm}$ are vector spherical harmonics in the cylindrical coordinates ($r' = a\theta, \theta' = \phi, z' = r$). The earlier expression is the same as equation (1) in [Zhu and Rivera \(2002\)](#) by noting that the horizontal wavenumber in the cylindrical coordinates $k = \frac{\kappa}{a} \approx \frac{l+1/2}{a}$. This means that for high-frequency body waves with large horizontal wavenumbers, the waveforms in a spherical layered medium with the EEFT is similar to those in the horizontal layered medium except the geometric spreading factor $\sqrt{\frac{\theta}{\sin \theta}}$ due to sphericity. Figure 2 shows seismograms calculated using sFK and f - k using the hypothetical event used in Figure 1. The waveform shapes and amplitudes of the two methods agree with each for body waves but differ noticeably for the long-period surface waves. The cross-correlation coefficients between the sFK and f - k synthetic waveforms range ~ 0.78 – 0.95 for the P – SV components and ~ 0.83 – 0.94 for the SH components, which are significantly lower than the cross-correlation coefficients between the sFK and DSM synthetic waveforms. We further calculated sFK and f - k synthetic waveforms at higher frequency (~ 2 Hz), and computed the corresponding cross-correlation coefficients. Results show that the sFK and f - k synthetic waveforms have lower cross-correlation coefficients at large epicentral distances, especially for the P – SV waveforms (Fig. S2).

Figure 2. Comparison of sFK (black) with frequency–wavenumber (f - k) (red) seismograms from 10° to 100° from a 100 km deep earthquake in the IASPEI91 model. The f - k synthetics were calculated using the traditional earth flattening transformation (EFT). The source parameters, epicentral distances of the records, and the applied band-pass filter are the same as in Figure 1. See caption of Figure 1 for detailed explanations of the denotations.

In horizontal layered media, the $4 \times 4\mathbf{M}$ matrix of the P – SV system is degenerated at $\omega = 0$, so the static displacement solution has to be derived differently from the dynamic solution ([Zhu and Rivera, 2002](#)). However, in spherical layered media after the EEFT, $\mathbf{M}$ is no longer degenerated at $\omega = 0$ (see equations A1–A3), and the static displacement solution can be obtained from the dynamic displacement solution directly by setting $\omega = 0$. [Pan et al. \(2015\)](#) derived static propagator matrix and displacement solution for a transversely isotropic spherical earth model using a similar approach as in this study but with a different density flattening transformation. Our static displacement solution can be served as an alternative solution to that of [Pan et al. \(2015\)](#).

In summary, we derived the propagator matrices and source terms for a spherical multilayered Earth model using the EEFT. We found that their differences from their counterparts in

horizontal layered media are inversely proportional to K and K^2 . The surface displacement produced by a point source can be obtained in a similar form as in horizontal layered media. In addition, all the source terms in a spherical layered model have a source–depth-dependent scaling factor, which differs from in a horizontal layered model by up to 6% for deep earthquakes. Computation of theoretical seismograms was implemented using the stable and fast generalized reflection and transmission coefficients method. Numerical tests show that our formulae and implementation are correct and efficient for computing full-wave seismograms, including the permanent displacements, at teleseismic distances up to 100° . The computer program package sFK has been seamlessly integrated into the popular f - k package of horizontal layered model by Zhu and Rivera (2002). Because the generalized reflection and transmission method is used, individual seismic phases can be isolated and analyzed semi-analytically (e.g., Kennett, 1983). Furthermore, our analytic expression of displacement in terms of the propagator matrices and source terms can be used to derive analytic derivatives of seismograms for full-wave waveform inversion (e.g., Hu and Zhu, 2017, 2018).

Data and Resources

The source code of sFK is available at <https://github.com/ShaoqianHu/sfk> (last accessed on August 2023). The supplemental material includes Figures S1 and S2. No data are used in this study.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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Appendix A

The propagator matrices for the spheroidal problem

The J matrix is as follows:

$$J = \begin{pmatrix} -v_\alpha & 0 & 0 & 0 \\ 0 & -v_\beta & 0 & 0 \\ 0 & 0 & v_\alpha & 0 \\ 0 & 0 & 0 & v_\beta \end{pmatrix}, \quad (A1)$$

in which

$$v_\alpha^2 = K^2 - \frac{(a\omega)^2}{\alpha^2} + \frac{25}{4} - \frac{8\beta^2}{\alpha^2}, \quad (A2)$$

$$v_\beta^2 = K^2 - \frac{(a\omega)^2}{\beta^2} - \frac{7}{4}. \quad (A3)$$

$$E = \begin{pmatrix} -1 & -\frac{v_\beta}{K} + \frac{5}{2K} & 1 & \frac{v_\beta}{K} + \frac{5}{2K} \\ \frac{v_\alpha}{K} + \frac{5}{2K} & 1 & \frac{v_\alpha}{K} - \frac{5}{2K} & 1 \\ -2\mu\gamma_1 - \frac{4\mu v_\alpha}{K^2} - \frac{2\mu}{K^2} & -\frac{2\mu v_\beta}{K} + \frac{\mu}{K} & 2\mu\gamma_1 - \frac{4\mu v_\alpha}{K^2} + \frac{2\mu}{K^2} & \frac{2\mu v_\beta}{K} + \frac{\mu}{K} \\ \frac{2\mu v_\alpha}{K} + \frac{3\mu}{K} & 2\mu\gamma_1 - \frac{2\mu v_\beta}{K^2} - \frac{3\mu}{K^2} & \frac{2\mu v_\alpha}{K} - \frac{3\mu}{K} & 2\mu\gamma_1 + \frac{2\mu v_\beta}{K^2} - \frac{3\mu}{K^2} \end{pmatrix}, \quad (A4)$$

$$E^{-1} = \frac{\eta}{2} \begin{pmatrix} -1 + \frac{3}{2v_\alpha} & -\gamma_1 \frac{K}{v_\alpha} + \frac{2}{K} - \frac{1}{Kv_\alpha} \frac{1}{2\mu} \left(1 - \frac{5}{2v_\alpha}\right) & \frac{K}{2\mu v_\alpha} \\ \gamma_1 \frac{K}{v_\beta} + \frac{1}{K} - \frac{3}{2Kv_\beta} & 1 + \frac{1}{2v_\beta} & -\frac{K}{2\mu v_\beta} & -\frac{1}{2\mu} \left(1 + \frac{5}{2v_\beta}\right) \\ 1 + \frac{3}{2v_\alpha} & -\gamma_1 \frac{K}{v_\alpha} - \frac{2}{K} - \frac{1}{Kv_\alpha} - \frac{1}{2\mu} \left(1 + \frac{5}{2v_\alpha}\right) & \frac{K}{2\mu v_\alpha} \\ -\gamma_1 \frac{K}{v_\beta} + \frac{1}{K} + \frac{3}{2Kv_\beta} & 1 - \frac{1}{2v_\beta} & \frac{K}{2\mu v_\beta} & -\frac{1}{2\mu} \left(1 - \frac{5}{2v_\beta}\right) \end{pmatrix}, \quad (A5)$$

in which

$$\frac{1}{\eta} = \frac{1}{\gamma} + \frac{4}{K^2}, \quad (A6)$$

$$\gamma = \frac{2K^2\beta^2}{(a\omega)^2}, \quad (A7)$$

and

$$\gamma_1 = 1 - \frac{1}{\gamma}. \quad (A8)$$

The 4×4 propagator matrix $\mathbf{a}$ is given as follows:

$$\frac{1}{\eta}a_{11} = C_\alpha - \gamma_1 C_\beta - \frac{3(Y_\alpha - Y_\beta)}{2K} - \frac{Y_\beta}{2K\eta} + \frac{4}{K^2}C_\beta, \quad (A9)$$

$$\frac{1}{\eta}a_{12} = \gamma_1 Y_\alpha - X_\beta - \frac{2(C_\alpha - C_\beta)}{K} + \frac{1}{K^2}Y_\alpha + \frac{5}{4K^2}Y_\beta, \quad (A10)$$

$$\frac{1}{\eta}a_{13} = \frac{C_\beta - C_\alpha}{2\mu} + \frac{5(Y_\alpha - Y_\beta)}{4\mu K}, \quad (A11)$$

$$\frac{1}{\eta}a_{14} = \frac{X_\beta - Y_\alpha}{2\mu} - \frac{25}{8K^2\mu}Y_\beta, \quad (A12)$$

$$\frac{1}{\eta}a_{21} = \gamma_1 Y_\beta - X_\alpha - \frac{(C_\alpha - C_\beta)}{K} - \frac{3}{2K^2}Y_\beta + \frac{15}{4K^2}Y_\alpha, \quad (A13)$$

$$\frac{1}{\eta}a_{22} = C_\beta - \gamma_1 C_\alpha - \frac{5}{2K}\left(\gamma_1 + \frac{1}{K^2}\right)Y_\alpha + \frac{2}{K}X_\alpha + \frac{1}{2K}Y_\beta + \frac{4}{K^2}C_\alpha, \quad (A14)$$

$$\frac{1}{\eta}a_{23} = \frac{X_\alpha - Y_\beta}{2\mu} - \frac{25}{8K^2\mu}Y_\alpha, \quad (A15)$$

$$\frac{1}{\eta}a_{24} = \frac{C_\alpha - C_\beta}{2\mu} + \frac{5(Y_\alpha - Y_\beta)}{4K\mu}, \quad (A16)$$

$$\begin{aligned} \frac{1}{\eta}a_{31} &= 2\mu\gamma_1(C_\alpha - C_\beta) + \frac{4\mu}{K}X_\alpha - \frac{3\mu}{K}X_\beta - \frac{3\mu}{K}\left(\gamma_1 + \frac{1}{K^2}\right)Y_\alpha \\ &+ \frac{\mu}{K}\left(3\gamma_1 - 1 - \frac{13}{4K^2}\right)Y_\beta - \frac{4\mu}{K^2}(C_\alpha - C_\beta), \end{aligned} \quad (A17)$$

$$\frac{1}{\eta}a_{32} = 2\mu(\gamma_1^2 Y_\alpha - X_\beta) + \frac{\mu}{2K^2}Y_\beta - \frac{8\mu}{K^2}X_\alpha + \frac{2\mu}{K^2}\left(2\gamma_1 + \frac{1}{K^2}\right)Y_\alpha, \quad (A18)$$

$$\frac{1}{\eta}a_{33} = C_\beta - \gamma_1 C_\alpha + \frac{5}{2K}\left(\gamma_1 + \frac{1}{K^2}\right)Y_\alpha - \frac{2}{K}X_\alpha - \frac{1}{2K}Y_\beta + \frac{4}{K^2}C_\alpha, \quad (A19)$$

$$\frac{1}{\eta}a_{34} = X_\beta - \gamma_1 Y_\alpha - \frac{2}{K}(C_\alpha - C_\beta) - \frac{1}{K^2}Y_\alpha - \frac{5}{4K^2}Y_\beta, \quad (A20)$$

$$\frac{1}{\eta}a_{41} = 2\mu(\gamma_1^2 Y_\beta - X_\alpha) + \frac{9\mu}{2K^2}Y_\alpha + \frac{2\mu}{K^2}\left(-5\gamma_1 + 1 + \frac{4}{K^2}\right)Y_\beta, \quad (A21)$$

$$\begin{aligned} \frac{1}{\eta}a_{42} &= -2\mu\gamma_1(C_\alpha - C_\beta) + \frac{4\mu}{K^2}(C_\alpha - C_\beta) - \frac{3\mu}{K}\left(\gamma_1 + \frac{1}{K^2}\right)Y_\alpha \\ &+ \frac{4\mu}{K}X_\alpha + \frac{\mu}{2K}\left(1 - \frac{5}{4K^2}\right)Y_\beta - \frac{3\mu}{2K}X_\beta, \end{aligned} \quad (A22)$$

$$\frac{1}{\eta}a_{43} = X_\alpha - \gamma_1 Y_\beta - \frac{1}{K}(C_\alpha - C_\beta) - \frac{15}{4K^2}Y_\alpha + \frac{3}{2K^2}Y_\beta, \quad (A23)$$

$$\frac{1}{\eta}a_{44} = C_\alpha - \gamma_1 C_\beta + \frac{3}{2K}(Y_\alpha - Y_\beta) + \frac{1}{2K\eta}Y_\beta + \frac{4}{K^2}C_\beta. \quad (A24)$$

Appendix B

The toroidal problem

The $\mathbf{M}$ matrix is as follows:

$$\mathbf{M} = K \begin{pmatrix} \frac{1}{2K} & \frac{1}{\mu} \\ \mu - \rho\left(\frac{a\omega}{K}\right)^2 - \frac{2\mu}{K^2} & -\frac{1}{2K} \end{pmatrix}. \quad (B1)$$

Its Jordan decomposition is as follows:

$$\mathbf{J} = \begin{pmatrix} -v_\beta & 0 \\ 0 & v_\beta \end{pmatrix}, \quad (B2)$$

$$\mathbf{E} = \begin{pmatrix} -1 & 1 \\ \frac{\mu(2\nu_\beta+1)}{2K} & \frac{\mu(2\nu_\beta-1)}{2K} \end{pmatrix},$$

and

$$\mathbf{E}^{-1} = \frac{1}{2} \begin{pmatrix} -1 + \frac{1}{2\nu_\beta} & \frac{K}{\mu\nu_\beta} \\ 1 + \frac{1}{2\nu_\beta} & \frac{K}{\mu\nu_\beta} \end{pmatrix}.$$

(B3) The corresponding propagator matrix is as follows:

$$\mathbf{a} = \begin{pmatrix} C_\beta - \frac{1}{2K} Y_\beta & -\frac{Y_\beta}{\mu} \\ -\mu X_\beta + \frac{\mu}{4K^2} Y_\beta & C_\beta + \frac{1}{2K} Y_\beta \end{pmatrix}. \quad (\text{B5})$$

(B4)

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